

A

Minor Project Synopsis on

AI-Based Dark Data Mining System

Submitted to



RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL (M.P)

In Partial fulfillment for the award of degree

of

BACHELOR OF TECHNOLOGY

IN

Computer Science Engineering

By

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Jul-Dec-2025



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DECLARATION

We hereby declare that the work, which is being presented in this dissertation entitled “**AI-Based Dark Data Mining System**” in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science Engineering**, is an authentic record of work carried out by us.

The matter embodied in this report has not been submitted by us for the award of any other degree.

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CERTIFICATE

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Technology in during the academic year

2025-26

PROJECT GUIDE

HEAD OF THE DEPARTMENT

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ABSTRACT

Enterprises generate massive volumes of data daily in the form of emails, PDFs, scanned documents, system logs, chat records, and multimedia files. A large portion of this data remains unstructured and unused, commonly referred to as *dark data*. Dark data often contains valuable business insights as well as sensitive information, which, if left unanalyzed, can lead to missed opportunities, increased storage costs, and compliance risks.

This project presents an **AI-Based Dark Data Mining System** designed to extract meaningful information from unstructured enterprise data. The system performs data ingestion from multiple sources, converts nontextual data into analyzable formats using OCR and speech-to-text techniques, and applies advanced AI and Natural Language Processing (NLP) methods. Key functionalities include document classification, named entity recognition for identifying sensitive data, sentiment analysis, summarization, and anomaly detection.

The extracted insights are visualized through an interactive dashboard to support informed decision-making. The proposed system helps organizations reduce risks, optimize data storage, and unlock hidden business value, thereby transforming dark data into actionable intelligence.



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ACKNOWLEDGEMENT

We take this opportunity to express our heartfelt gratitude to everyone who has supported us throughout the journey of completing our minor project.

First and foremost, we would like to thank the Head of the Department,

Dr. Richa Gupta for their encouragement, guidance, and providing us with all the necessary facilities to carry out this project.

We extend our sincere gratitude to our Project Coordinator, **Ms. Megha Birthare**, for their continuous support, timely advice, and valuable feedback. We are deeply indebted to our Project Guide, **Mr. Amit Kanungo**, for their expert guidance, valuable suggestions, and support throughout this project.

Thank you all for making this project a meaningful and enriching experience.

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CHAPTER 1

INTRODUCTION

1.1 Background

In today's digital era, organizations generate a massive amount of data every day through emails, documents, system logs, chat messages, scanned files, and multimedia content. A large portion of this data remains unused and unstructured, commonly referred to as **dark data**. Although this data is stored for long periods, it is rarely analyzed due to the lack of proper tools and techniques.

Dark data may contain valuable business insights, security risks, compliance-related information, or hidden trends that can support better decision-making. However, traditional data management systems focus mainly on structured data and fail to extract knowledge from unstructured sources.

With advancements in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP), it has become possible to analyze dark data efficiently. AI-based systems can extract text from documents, detect sensitive information, identify patterns, and generate meaningful insights. This project focuses on developing an **AI-Based Dark Data Mining System** that helps enterprises convert unused data into useful and actionable intelligence.

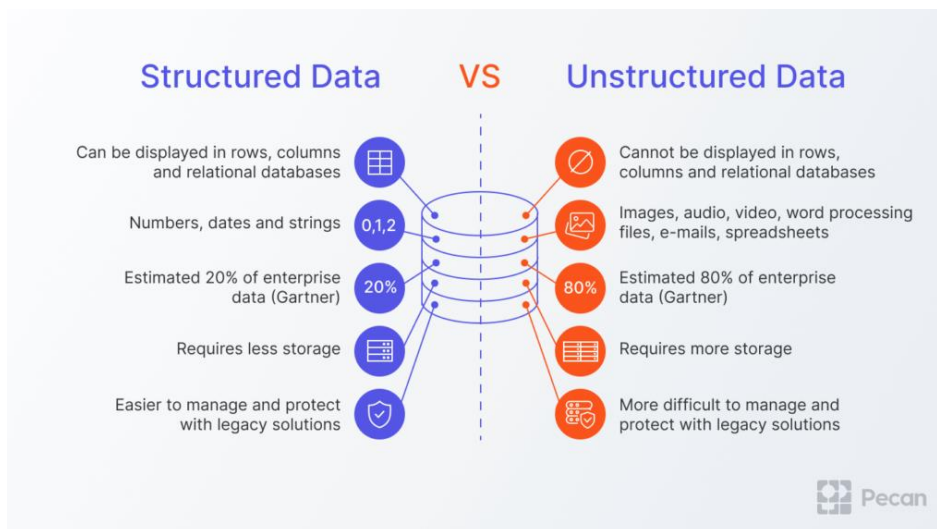


Figure 1.1: Illustration of dark data where large volumes of unstructured enterprise data remain unused and require AI-based analysis for extracting actionable insights.

1.2 Problem Statement

Organizations store huge volumes of unstructured data such as emails, PDFs, scanned documents, system logs, and chat records. Due to the lack of automated analysis tools, this data remains unused and unmanaged. As a result, enterprises face problems such as increased storage costs, undetected security risks, compliance violations, and missed business opportunities.

Manual analysis of such large datasets is time-consuming, error-prone, and impractical. Existing systems do not provide a centralized and intelligent solution to process dark data efficiently. Therefore, there is a need for an automated AI-driven system that can analyze unstructured data, detect sensitive information, and generate valuable insights.

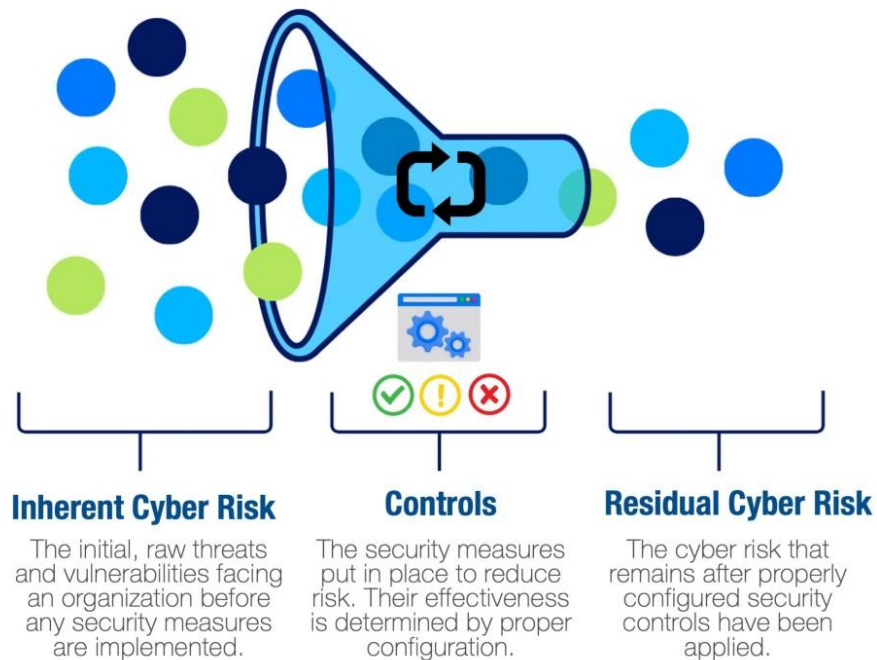


Figure 1.2: Representation of challenges caused by unmanaged unstructured data, including security risks and compliance issues.

1.3 Purpose/Objective

The main purpose of this project is to design and develop an **AI-Based Dark Data Mining System** that extracts useful information from unstructured enterprise data.

Specific Objectives:

- To collect and process unstructured data from multiple sources
- To extract text using OCR and speech-to-text techniques
- To analyze data using AI and NLP methods
- To detect sensitive information such as PII and confidential data
- To generate summaries, trends, and insights
- To provide an interactive dashboard for visualization and reporting

1.4 Solution Approach

The proposed system follows a modular and scalable architecture. Data is first collected from various sources and converted into readable formats. AI and NLP techniques are then applied to classify documents, extract entities, analyze sentiment, and identify anomalies.

The system uses a layered architecture consisting of data ingestion, preprocessing, AI analysis, and visualization layers. This approach ensures better performance, easy maintenance, and future scalability. The final insights are displayed through a dashboard and can also be accessed via APIs.

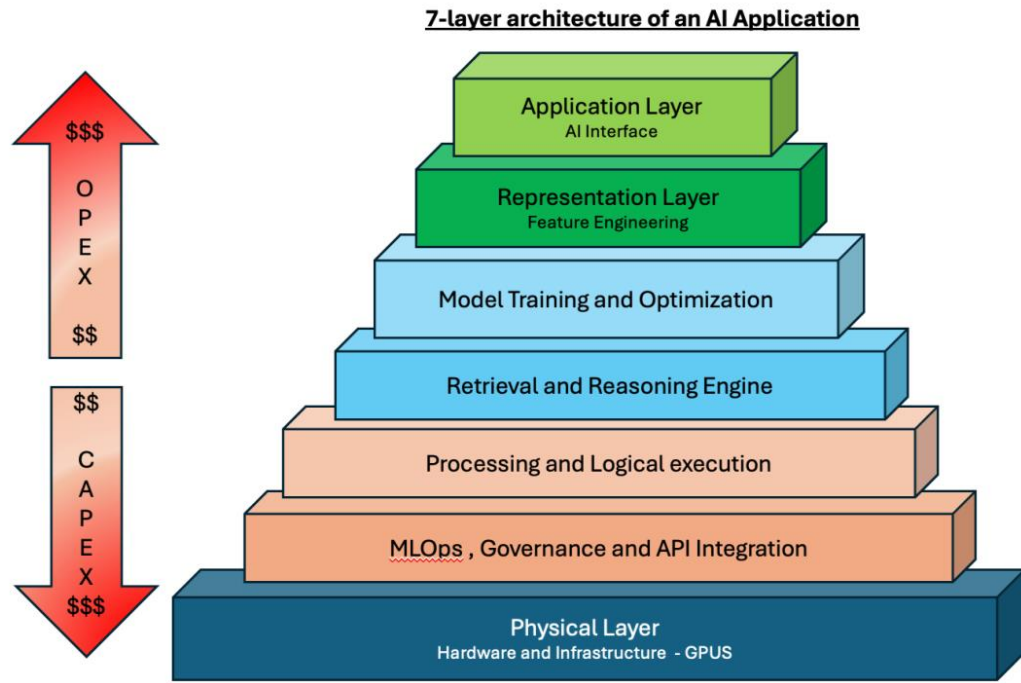


Figure 1.4: Layered architecture of the AI-Based Dark Data Mining System including ingestion, preprocessing, AI analysis, and visualization

1.5 Scope of the Project

The scope of this project includes the design and development of an AI-powered system to analyze enterprise dark data.

Key Features:

1. Data Ingestion:

- Supports emails, PDFs, scanned documents, logs, and transcripts

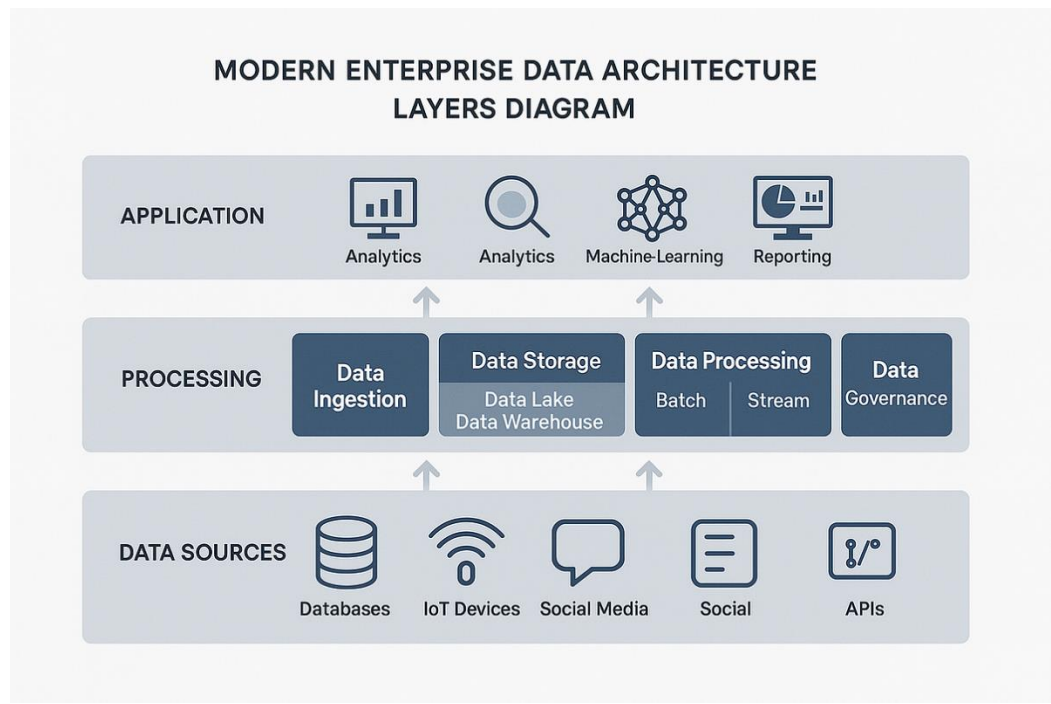


Figure 1.5.1: Data ingestion module collecting emails, documents, logs, and multimedia files for centralized processing.

2. AI & NLP Processing:

- Text classification
- Named Entity Recognition (NER)
- Sentiment analysis
- Document summarization

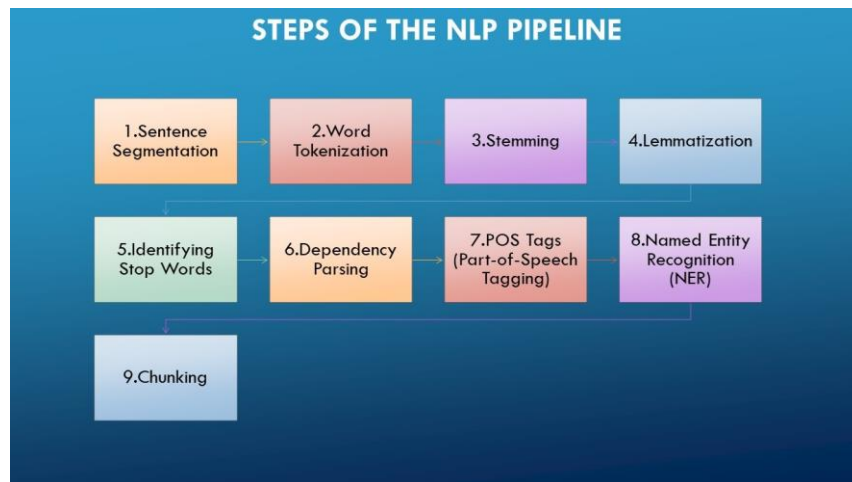


Figure 1.5.2: NLP processing pipeline performing classification, named entity recognition, sentiment analysis, and summarization.

3. Risk Detection:

- Identification of sensitive and confidential information
- Compliance and security alerts

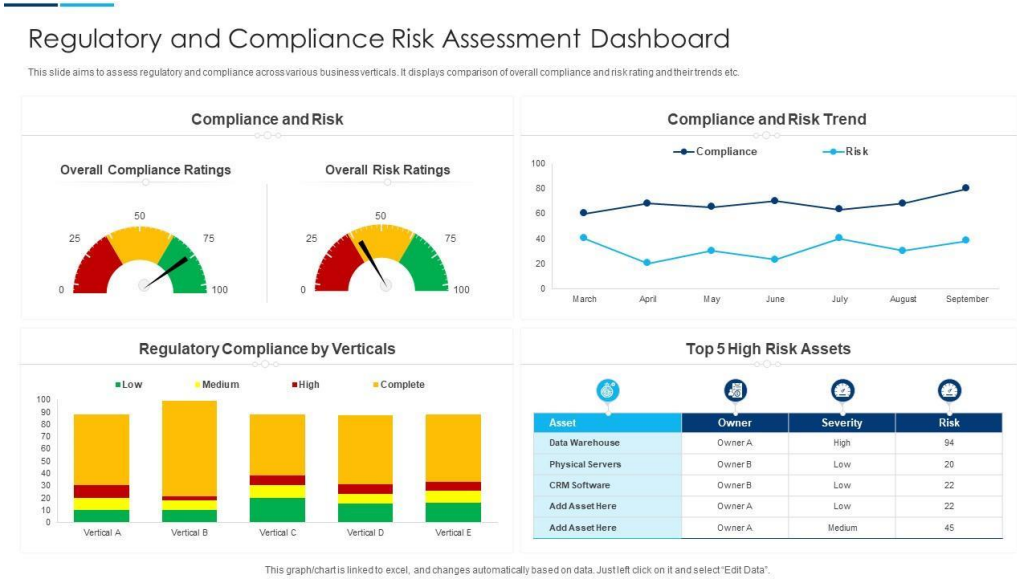


Figure 1.5.3: AI-based detection of sensitive information such as PII and confidential enterprise data.

4. Visualization & Reporting:

- Dashboards showing trends, risks, and insights
- Exportable reports

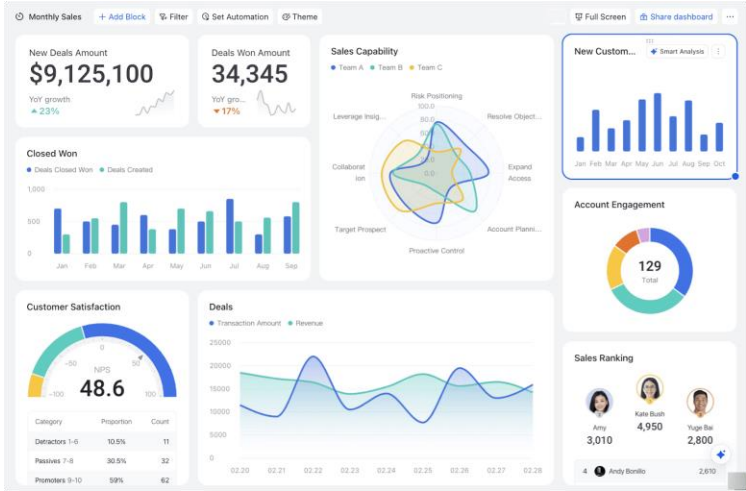


Figure 1.5.4: Interactive dashboard displaying insights, trends, anomalies, and risk alerts.

5. Admin Features:

- Data monitoring
- Analysis control

- Report generation

1.6 Existing Systems & Limitations

Traditional enterprise data management systems mainly focus on structured databases and ignore unstructured data. Existing analytics tools require manual preprocessing and lack intelligence for automated analysis.

Most systems are unable to detect sensitive information hidden inside documents or logs. They also fail to provide meaningful insights from historical data, leading to inefficiencies and increased operational risks.

1.6.1 Limitations of Current Systems

- Manual analysis of unstructured data
- Lack of AI-driven automation
- High storage costs for unused data
- Poor risk and compliance detection
- No centralized dashboard for insights

CHAPTER 2

METHODOLOGY

2.1 Proposed Methodology

The **Agile Development Model** is adopted for this project to allow flexibility and continuous improvement. Agile supports iterative development, frequent testing, and regular feedback, ensuring that the system meets user and enterprise requirements effectively.

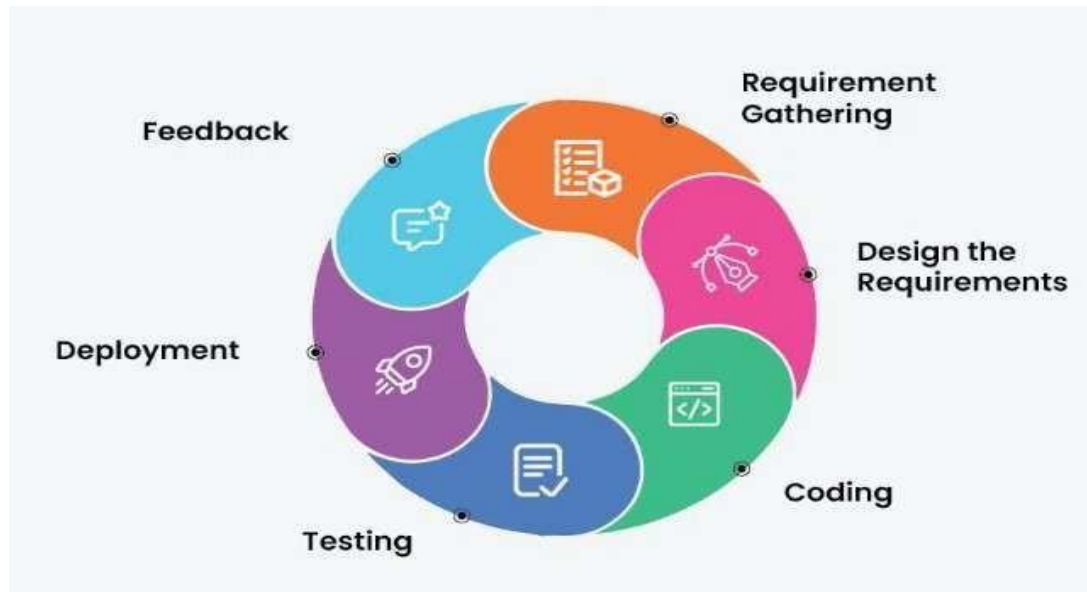


Fig-2.1: Agile Development Model

2.2 Process Model Adopted

The **Scrum Framework** is used to manage development activities. The project is divided into short sprints involving data ingestion, AI model integration, testing, and visualization development.

- Sprint Planning
- Development & Integration
- Testing & Review
- Retrospective

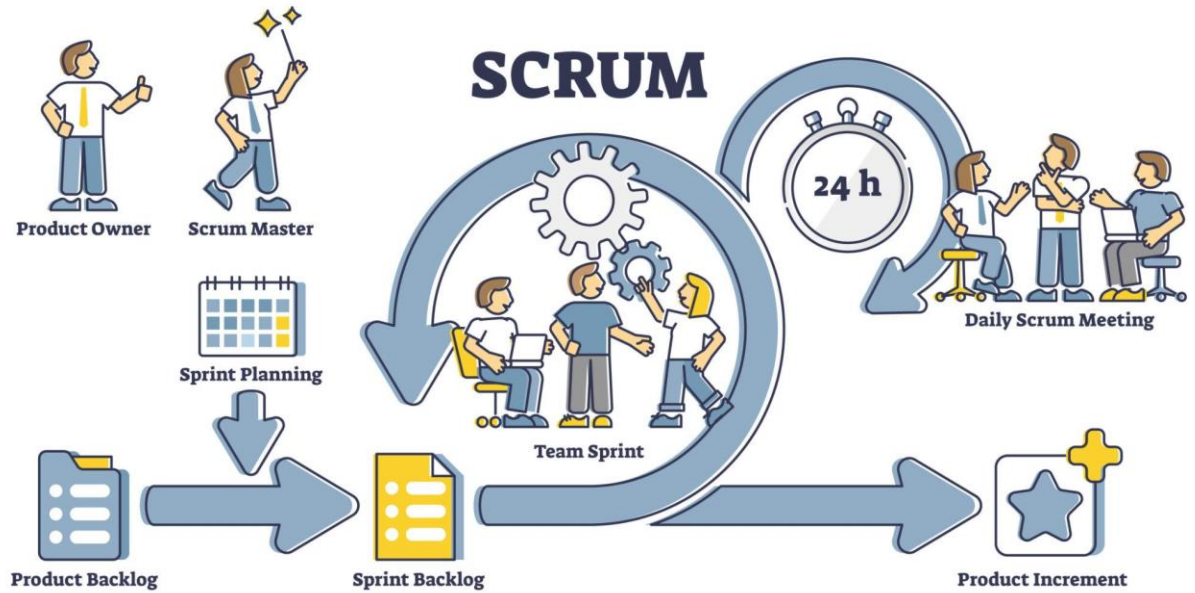


Figure 2.2: Scrum framework used to manage sprint planning, development, review, and retrospective activities.

2.3 Planning and Scheduling

1. Requirement Analysis

In this phase, the project requirements are carefully analyzed. The types of dark data sources, system features, user roles, and functional needs are identified. This phase helps in understanding what the system should do and defines the overall scope of the project.

2. System Design

During this phase, the architecture of the AI-Based Dark Data Mining System is designed. This includes designing data flow, system modules, database structure, and interaction between components such as data ingestion, AI processing, and dashboard visualization.

3. AI Model Integration

This phase focuses on implementing and integrating AI and NLP models. OCR is used for document text extraction, ASR for audio transcription, and NLP techniques for classification, entity extraction, sentiment analysis, and summarization. Testing and tuning of models are also performed.

4. Dashboard Development

In this stage, an interactive dashboard is developed to display insights, trends, risk alerts, and reports. The dashboard is designed to be user-friendly and responsive, allowing administrators and decision-makers to easily interpret the analyzed data.

5. Testing & Documentation – 1 Week

The final phase involves testing the complete system to ensure accuracy, performance, and security. Any identified issues are fixed. Proper documentation is prepared, including system design, user guides, and project reports.

CHAPTER 3

REQUIREMENTS AND ANALYSIS

3.1 Problem Definition

In today's digital environment, enterprises generate and store huge amounts of data in the form of documents, emails, audio files, videos, logs, and scanned records. Most of this data is **unstructured**, meaning it cannot be easily analyzed using traditional database or reporting systems. As a result, a large portion of valuable information remains unused and is referred to as **dark data**.

Due to the lack of automated AI-based systems, organizations find it difficult to extract meaningful insights from this dark data. Important information related to customer behavior, operational risks, compliance issues, and business trends often remains hidden. Manual analysis of such large datasets is time-consuming, expensive, and prone to errors.

The main challenge is to design an efficient system that can automatically collect, process, and analyze unstructured data using artificial intelligence techniques. The system should be able to extract useful information, identify patterns and risks, and convert raw dark data into **actionable intelligence** that supports better decision-making for enterprises.

3.2 Feasibility Study

A **feasibility study** is a detailed analysis conducted to determine whether a proposed system can be developed and implemented successfully. It evaluates the project from technical, economic, and operational perspectives to ensure that the system is practical, affordable, and usable.

In software projects, a feasibility study helps identify whether the required technologies are available, whether the project can be completed within budget and time constraints, and whether end users will be able to operate the system effectively.

For this project, the feasibility study confirms that the AI-Based Dark Data Mining System can be developed using existing technologies, implemented at low cost using open-source tools, and operated efficiently with minimal training.

3.2.1 Technical Feasibility

Technical feasibility focuses on whether the required technologies, tools, and technical skills are available to build the proposed system.

The AI-based Dark Data Mining System is developed using **Python**, which is widely used in artificial intelligence and data processing applications. Python supports powerful libraries such as **NLTK, spaCy, and Transformers** for Natural Language Processing (NLP), enabling efficient text classification, entity extraction, and summarization.

For processing scanned documents and images, **OCR tools** like Tesseract are used to convert image-based data into machine-readable text. Audio and video data can be converted into text using automatic speech recognition models. Cloud databases are used to store large volumes of data securely and efficiently.

All these technologies are **open-source, well-documented, and industry-proven**, which reduces development risk. The integration of AI models is supported by existing frameworks, making system development smooth and scalable. Therefore, the project is considered **technically feasible**.

3.2.2 Economic Feasibility

Economic feasibility evaluates whether the project can be developed within acceptable financial limits.

This project is designed to be **cost-effective** by using open-source tools and libraries, which eliminates licensing costs. Free-tier or low-cost cloud services are sufficient for data storage, model execution, and testing during development.

Since the project does not require specialized hardware or expensive software, the overall development cost remains minimal. The resources required are already available in most

academic environments. As a result, the project is financially affordable and suitable for a student-level minor project. Hence, the project is **economically feasible**.

3.2.3 Operational Feasibility

Operational feasibility examines whether the system can be easily used and accepted by its intended users.

The proposed system provides a **simple and intuitive dashboard** that allows administrators to upload data, monitor analysis progress, view insights, and generate reports. Most of the complex processing is handled automatically by the system, reducing manual effort.

Users do not require advanced technical knowledge to operate the system. Minimal training is sufficient to understand system functionality. The system improves efficiency by saving time and reducing human errors in data analysis. It can be easily integrated into existing organizational workflows, making daily operations smoother.

Therefore, the system is **operationally feasible** and practical for real-world enterprise usage.

3.3 Functional Requirements

Functional requirements specify **the core functionalities and services** provided by the AI-based Dark Data Mining System.

3.3.1 Upload and Ingest Dark Data

The system shall provide a secure and flexible mechanism for uploading and ingesting dark data from multiple sources. Users will be able to upload files such as emails, PDF documents, scanned images, system log files, chat histories, and audio or video recordings.

The ingestion module will validate file formats, check file size limits, and store the data in a structured repository. The system must handle bulk uploads efficiently to support enterprise-level data volumes. This requirement ensures that all unstructured data is successfully collected and made ready for further processing.

3.3.2 Convert Data Using OCR and ASR

Since dark data is mostly unstructured, the system shall convert it into machine-readable text format. Optical Character Recognition (OCR) will be used to extract text from scanned documents and images that do not contain selectable text.

For audio and video files, Automatic Speech Recognition (ASR) technology will be used to transcribe spoken content into text. This conversion step is critical, as AI and NLP models require textual input to perform analysis. The system shall ensure high accuracy during conversion to minimize information loss.

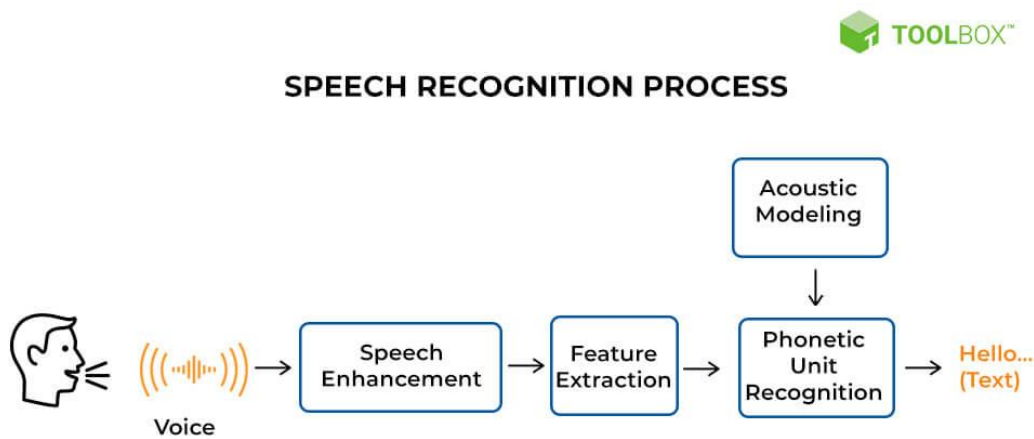


Figure 3.3.2: OCR and ASR modules converting scanned documents and audio files into machine-readable text.

3.3.3 Classify Documents

The system shall automatically classify documents based on their content using AI and Natural Language Processing techniques. Documents may be categorized into predefined classes such as finance, human resources, legal, confidential, technical, or general.

Classification helps enterprises organize massive volumes of data and retrieve relevant information quickly. This feature reduces manual effort and improves data management efficiency by grouping similar documents together.

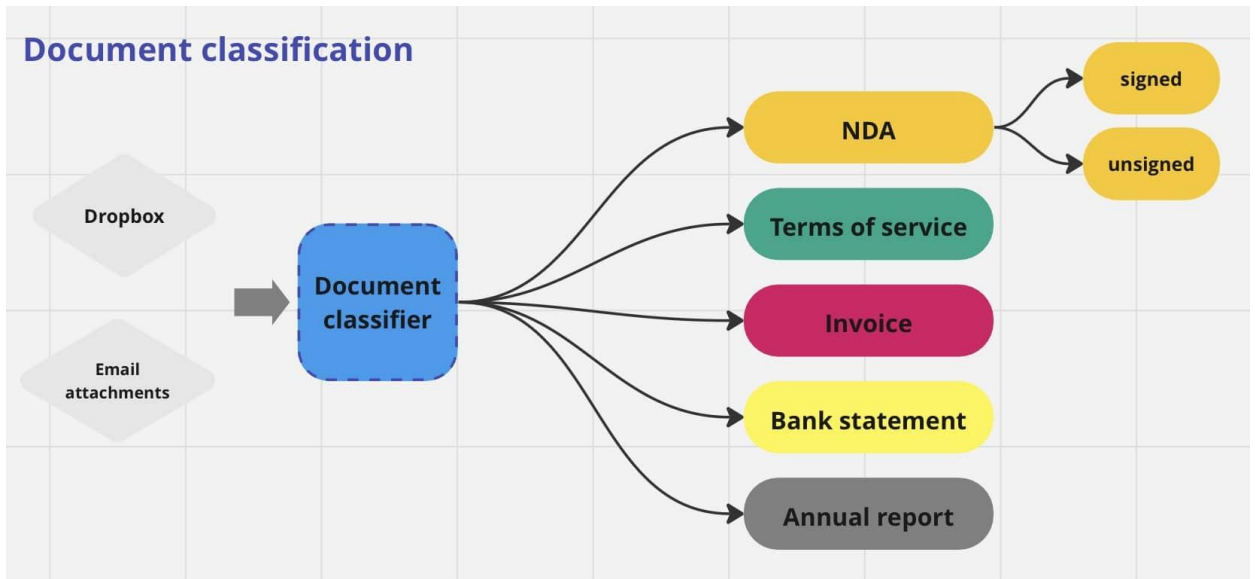


Figure 3.3.3: AI-based document classification system categorizing enterprise data into predefined classes.

3.3.4 Detect Sensitive Information

The system shall detect and identify sensitive information embedded within documents and logs. This includes Personally Identifiable Information (PII) such as names, addresses, phone numbers, Aadhaar numbers, bank details, and confidential business information.

Named Entity Recognition (NER) and pattern-matching techniques will be used to flag risky data. The system will generate alerts when sensitive content is detected, helping organizations reduce security threats and ensure compliance with data protection regulations like GDPR.

3.3.5 Generate Summaries and Reports

The system shall generate concise and meaningful summaries of lengthy documents using AI-based text summarization techniques. This allows users to understand the key points of large files without reading them entirely.

Additionally, the system shall generate analytical reports containing insights, trends, detected risks, and classified data statistics. Reports can be exported in standard formats such as PDF and CSV, making them suitable for audits, presentations, and decision-making.

3.3.6 Visualize Insights on Dashboard

The system shall provide an interactive and visual dashboard for presenting processed data and insights. Graphs, charts, tables, and alerts will be displayed to show trends, anomalies, sentiment analysis results, and risk indicators.

The dashboard will allow users to filter data based on time, category, or risk level. This visualization helps stakeholders easily interpret complex AI outputs and make informed business decisions.

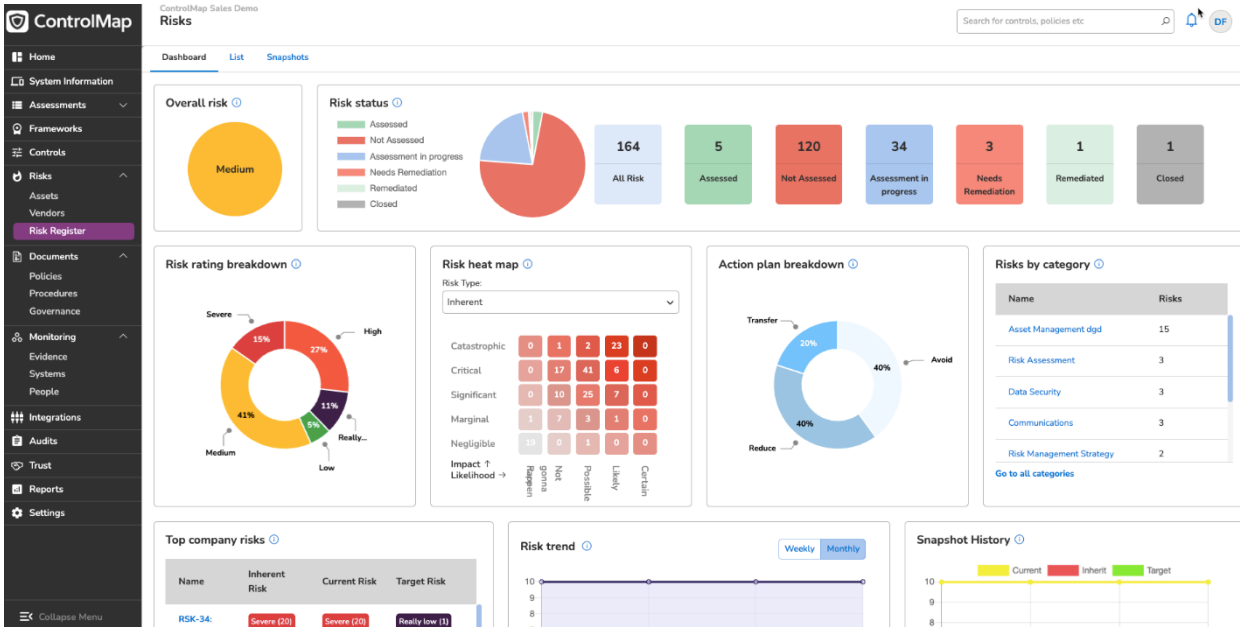


Figure 3.3.6: Visualization dashboard presenting processed insights, anomalies, and detected risks.

3.4 Non-Functional Requirements

Non-functional requirements define **performance, quality, and usability characteristics** of the system.

3.4.1 High Performance

The system shall process large volumes of dark data efficiently with minimal delay. AI model execution, document classification, and report generation should be optimized to ensure fast response times.

Batch processing and parallel execution techniques may be used to enhance performance. High performance ensures the system remains practical for enterprise-scale use.

3.4.2 Secure Data Handling

The system shall ensure complete data security at all stages, including upload, storage, processing, and reporting. Security mechanisms such as user authentication, role-based access control, and encryption will be implemented.

Sensitive data detected by the system shall be protected from unauthorized access. Secure handling is critical to maintain trust and comply with organizational and legal security standards.

3.4.3 Scalability

The system shall be scalable to accommodate growing data volumes and increasing numbers of users. Cloud-based infrastructure allows the system to expand storage capacity and processing power as required.

Scalability ensures that the system remains effective even as enterprise data continues to grow over time.

3.4.4 Reliability

The system shall operate reliably with minimal downtime. Proper error handling, logging, and recovery mechanisms will be implemented to handle failures gracefully.

Regular data backups and system monitoring will ensure that data is not lost and the system remains available when needed. High reliability is essential for enterprise-level applications.

3.4.5 User-Friendly Interface

The system shall provide an intuitive and easy-to-use interface. Users with limited technical knowledge should be able to upload data, view results, and generate reports without extensive training.

Clear navigation, well-designed dashboards, and informative messages will improve usability and user satisfaction.

3.5 Technical Requirements

Technical requirements describe the **tools, technologies, and platforms** required to develop and run the AI-based Dark Data Mining System.

3.5.1 Python

Python is the primary programming language used for developing the system. It is widely used in data science and artificial intelligence due to its simplicity, readability, and extensive library support. Python enables easy integration of AI models, data processing modules, and backend services.

Python also supports rapid development and testing, making it suitable for academic and enterprise-level projects. Its compatibility with web frameworks allows seamless connection between the backend processing and the dashboard interface.

3.5.2 NLP Libraries

Natural Language Processing (NLP) libraries are used to analyze and understand textual data extracted from dark data sources. Libraries such as NLTK, spaCy, and Transformers provide tools for text classification, named entity recognition, sentiment analysis, and summarization.

These libraries enable the system to extract meaningful insights from unstructured text data. NLP libraries play a crucial role in converting raw text into structured and actionable information.

3.5.3 OCR (Tesseract)

Optical Character Recognition (OCR) is required to extract text from scanned documents and image-based files. Tesseract OCR is an open-source and widely used OCR engine that supports multiple languages and document formats.

Tesseract converts image-based content into machine-readable text, allowing further AI and NLP processing. This technology is essential for handling scanned PDFs, handwritten forms, and image documents commonly found in enterprise dark data.

3.5.4 ASR (Whisper)

Automatic Speech Recognition (ASR) is used to convert audio and video content into text format. Whisper is an advanced ASR model capable of accurately transcribing speech from recordings, even in noisy environments.

By converting spoken data into text, Whisper enables analysis of customer calls, meeting recordings, and video content. This makes audio and video data searchable and analyzable using AI models.

3.5.5 MongoDB / Cloud Storage

MongoDB is used as a NoSQL database to store unstructured and semi-structured data efficiently. It supports flexible schema design, making it suitable for storing diverse data types such as documents, logs, and extracted text.

Cloud storage is used to store large files such as PDFs, images, and audio/video recordings. Using cloud services improves scalability, availability, and reliability while reducing infrastructure costs. Together, MongoDB and cloud storage ensure secure and efficient data management.

3.6 Software Design

3.6.1 Flowchart

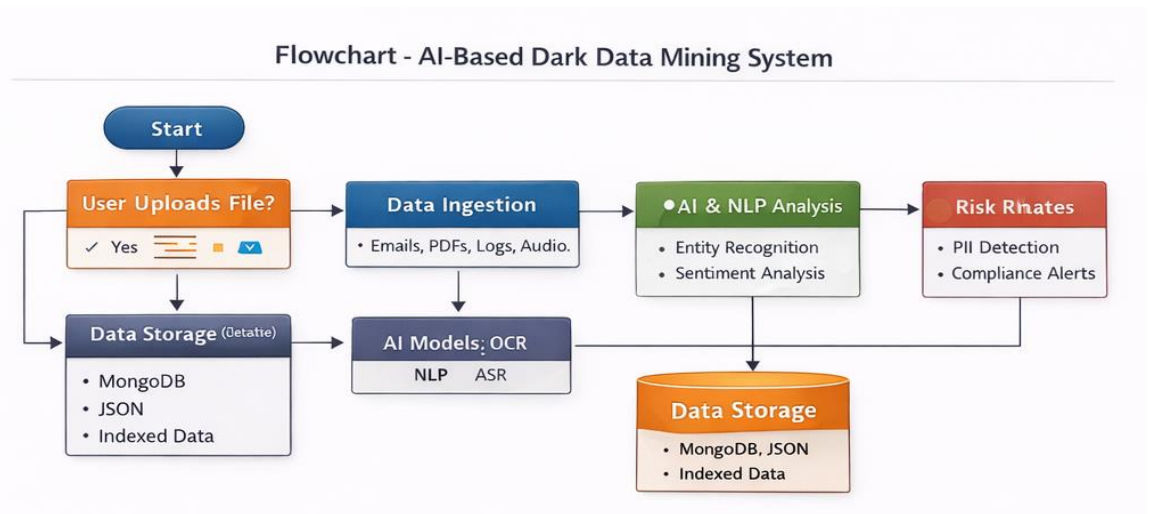
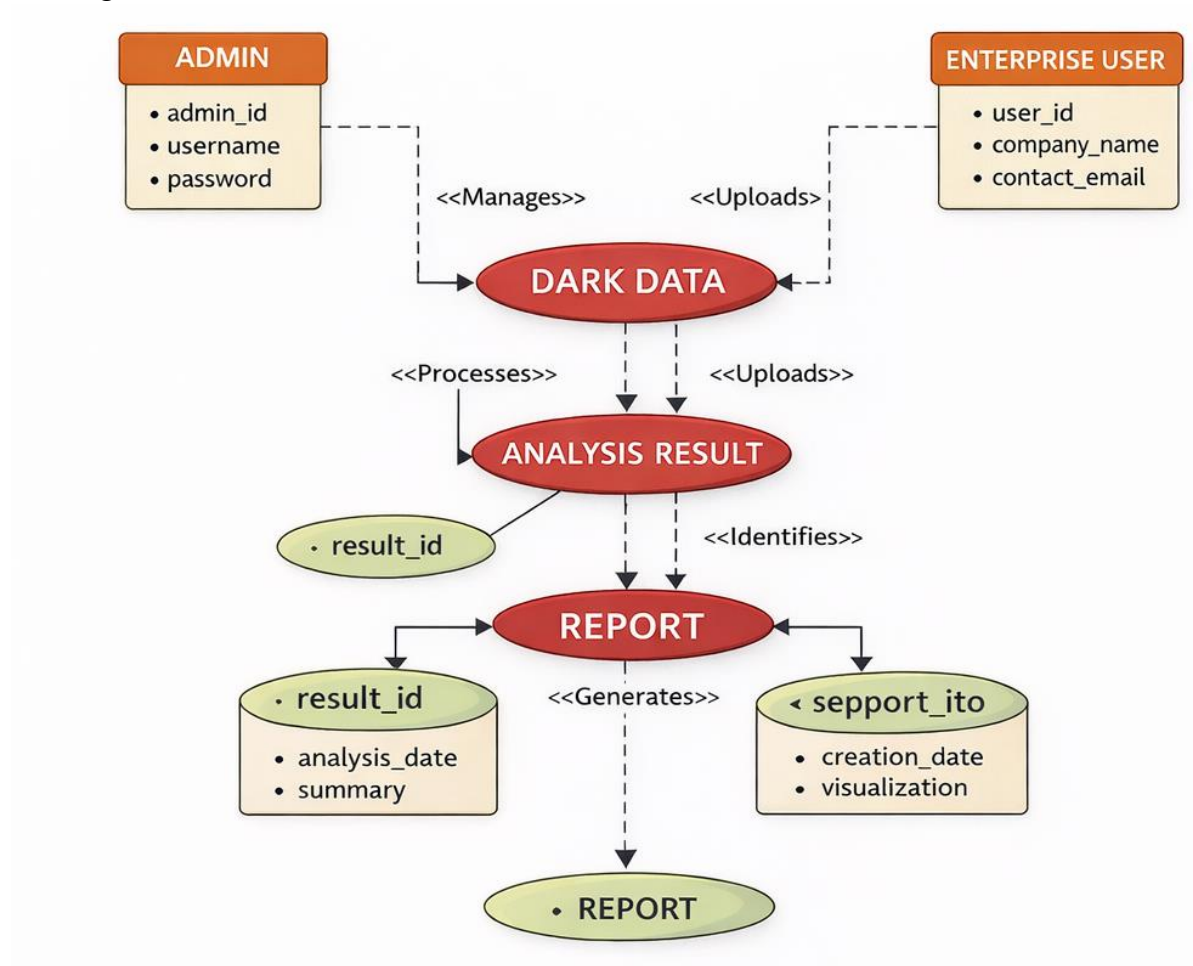


Fig 3.6.1: Overall Flow of AI-Based Dark Data Mining System

3.6.2 ER Diagram



CHAPTER 4

TECHNOLOGY STACK

4.1 Front-End Technology

HTML5, CSS3, JavaScript, React.js are used to build an interactive and responsive dashboard.

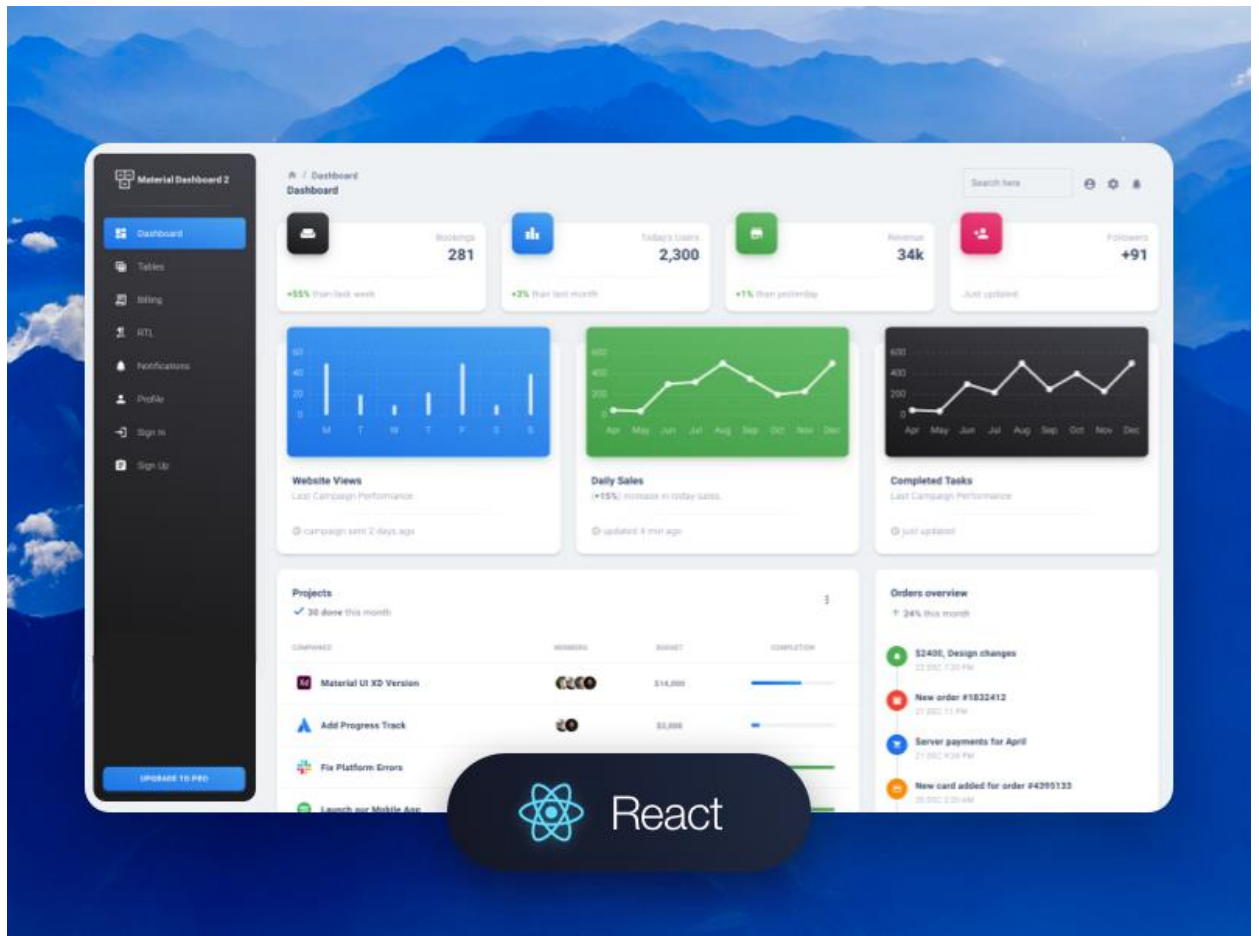


Figure 4.1: Front-end dashboard built using HTML, CSS, JavaScript, and React.js.

4.2 Back-End

Python and Flask are used for AI processing and API development.

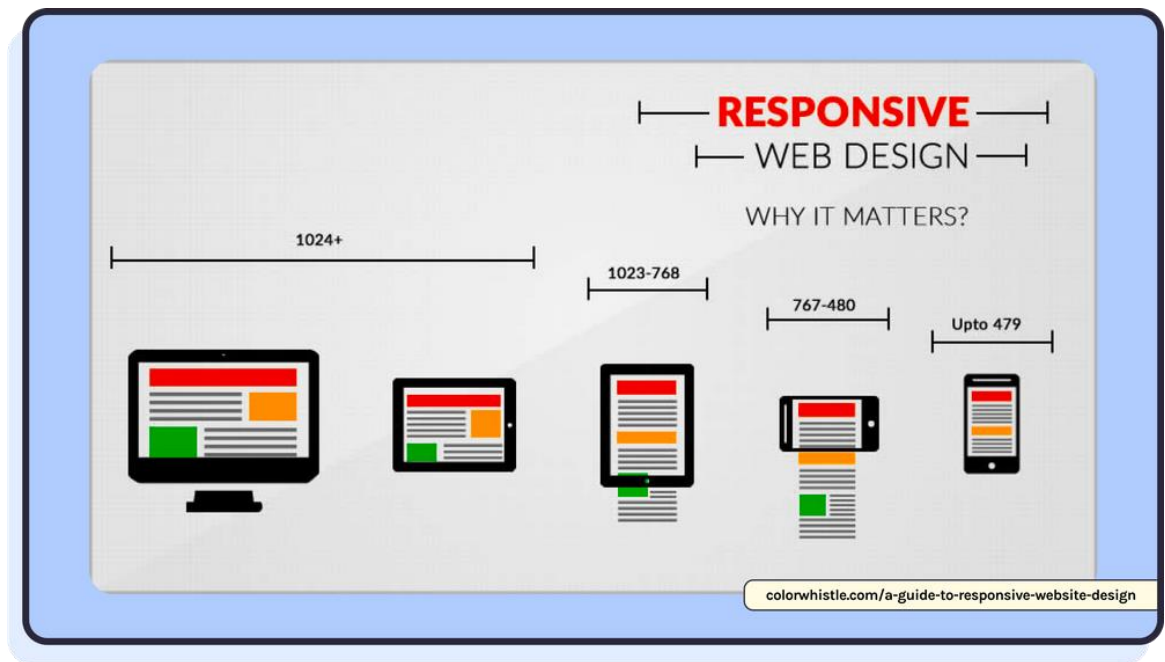


Figure 4.2: Backend architecture using Python and Flask for AI processing and API communication.

4.3 Database

MongoDB Atlas is used for scalable and secure storage of unstructured data and metadata.

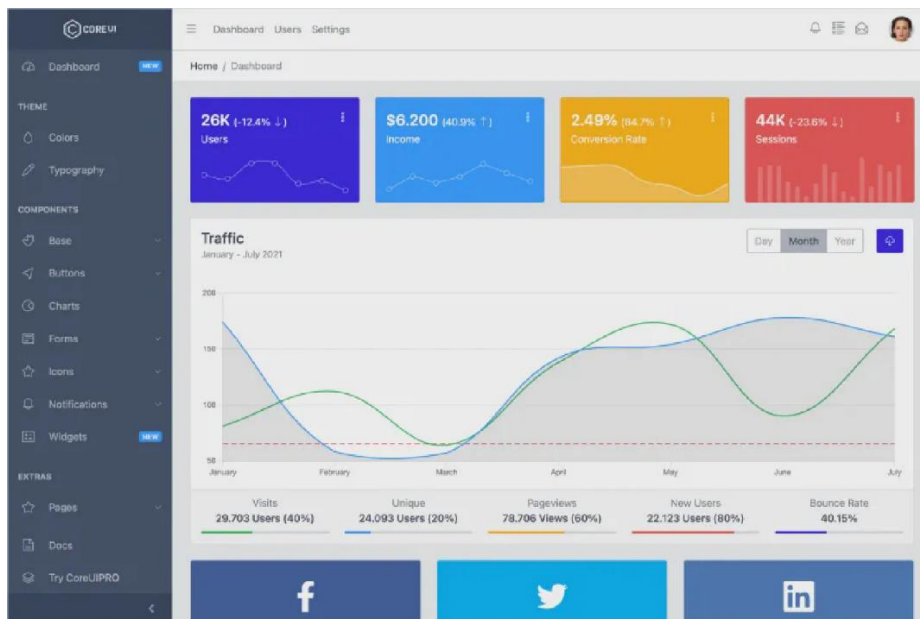


Figure 4.3: MongoDB and cloud storage architecture for handling unstructured enterprise data.

4.4 APIs & Tools

- **OCR – Tesseract**

Tesseract OCR is used to convert scanned documents and images into machine-readable text. It enables the system to extract useful information from PDFs and scanned files that cannot be processed directly.

- **ASR – Whisper**

Whisper is an automatic speech recognition tool used to convert audio and video files into text. This allows spoken content to be analyzed using NLP techniques.

- **GitHub – Version Control**

GitHub is used for version control and collaboration. It helps track code changes, manage project versions, and maintain a backup of the project source code.

CHAPTER 5

CONCLUSION LIMITATIONS AND FUTURE WORK

5.1 Conclusion

The AI-Based Dark Data Mining System successfully converts unstructured enterprise data into meaningful insights. The system improves data utilization, reduces risks, and supports informed decision-making.

5.2 Limitations

1. Limited to Batch Processing

The system processes data in batches, meaning files must be uploaded and analyzed together rather than continuously. This approach simplifies implementation and reduces system complexity, but it can cause delays in obtaining results. Batch processing is suitable for historical or archived data, but it is not ideal for applications that require instant insights or immediate decision-making.

2. No Real-Time Streaming Analysis

The current system does not support real-time streaming data analysis from live sources such as emails, logs, or sensors. Data must be manually uploaded before processing begins. This limits the system's ability to detect issues or patterns instantly. Implementing real-time streaming requires advanced infrastructure, which is beyond the scope of this project.

5.3 Future Work

1. Real-Time Data Analysis

In future versions, the system can be enhanced to support real-time data processing. This will allow continuous analysis of live data streams and provide instant insights. Real-time analysis can help organizations detect security threats, compliance risks, or important patterns immediately, enabling faster and more informed decision-making.



Figure 5.3.1: Future enhancement enabling real-time streaming analysis of enterprise data.

2. Advanced Predictive Analytics

Future improvements may include predictive analytics using machine learning and deep learning models. These models can analyze historical data to predict future trends, risks, or behaviors. Predictive analytics will enable the system to move beyond basic data analysis and support proactive planning and strategic decision-making.

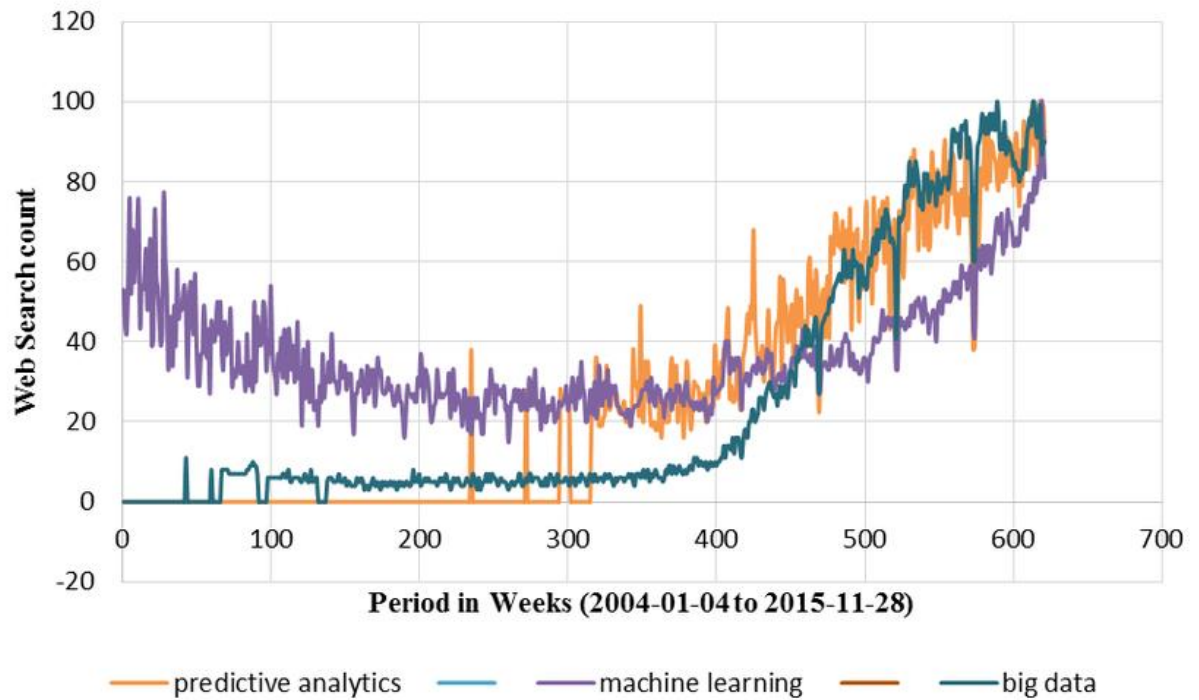


Figure 5.3.2: Predictive analytics models for forecasting trends and identifying future risks.

3. Cloud-Scale Deployment

The system can be deployed on cloud platforms to improve scalability, availability, and performance. Cloud deployment will allow the system to handle large volumes of data efficiently and support multiple users simultaneously. It also provides better reliability, automatic backups, and easier maintenance for enterprise-level applications.

REFERENCES

- **Research Papers on Dark Data**

Research papers on dark data provide in-depth knowledge about unstructured data, its challenges, and techniques used to extract valuable insights. These papers help in understanding real-world problems, existing solutions, and future research directions related to dark data mining.

- **AI & NLP Documentation**

AI and NLP documentation includes official guides and manuals of machine learning and natural language processing libraries. These resources explain algorithms, implementation methods, and best practices used for text analysis, classification, and information extraction.

- **OCR and ASR Tool Documentation**

OCR and ASR tool documentation provides detailed information on text extraction from images and speech-to-text conversion. These documents help in understanding tool configuration, accuracy improvement techniques, and integration methods within the system.

